Morphology Of Flowering Plants

INTRODUCTION

Katherine Esau

- She absolutely **dominated** in the field of '**Plant Biology**' even at the age of 99.
- She was elected to 'National Academy of Sciences' and became **6th woman** to receive this honour.
- She received '**National Medal of Science**' from President Bush in 1989.

Publications : → Her early publications depict that **curly top virus** spreads through phloem.
→ Published '**Plant Anatomy**' in 1954.
→ Published '**Anatomy of Seed plants**' in 1960.

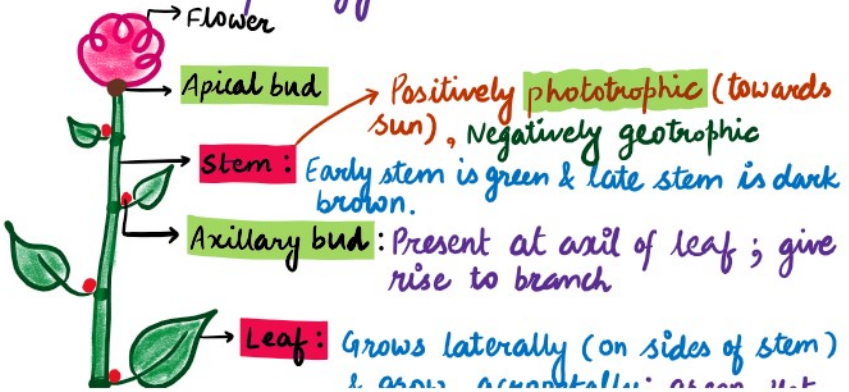
This was called '**Webster of Plant Biology**'.

MORPHOLOGY

External structure | Study

- Morphology is the study of external structures which may be :
  - Vegetative : like root, stem & leaves.
  - Reproductive : Flower
  - Products of sexual reproduction : Fruit and seed
- Adaptations of these organs for storage, climbing, protection etc are also studied under morphology.

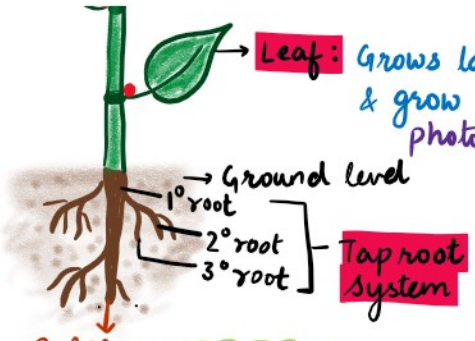
shoot system (above the ground) → Develop from plumule



Develop from plumule

Root System (underground)

Develops from Radicle



Leaf: Grows laterally (on sides of stem) & grow acropetally; green, flat, photosynthetic structure



Positively geotrophic
Negatively phototrophic

THE ROOT

TYPES OF ROOTS

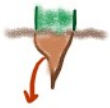
Tap roots



- 1° root and all its branches constitutes 'Tap root system'.

e.g., Most Dicots like Mustard.

Fibrous roots



short lived 1° root replaced by



Large no. of fibrous roots

e.g., in most monocots like - Wheat

Adventitious roots

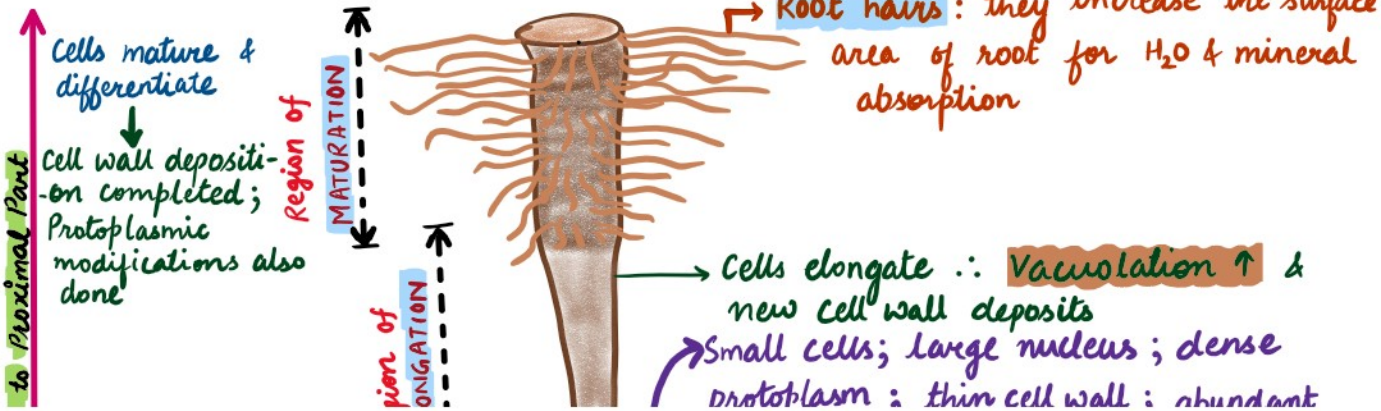
In some plants, root arise from part other than radicle

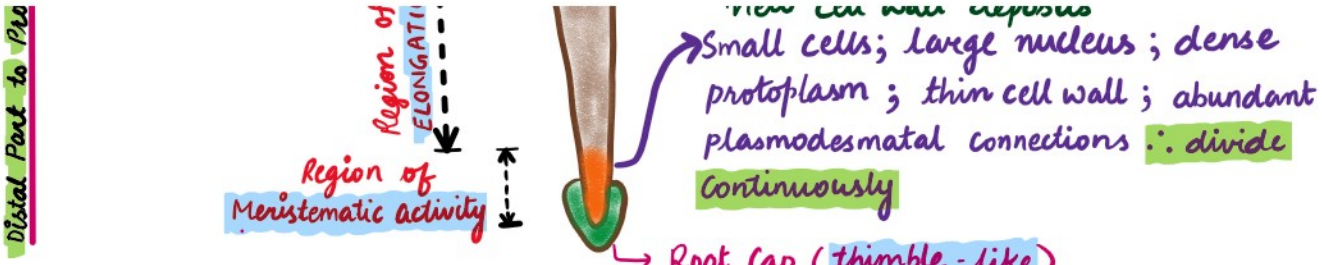
e.g., Grasses, Banyan tree, Monstera



Adventitious root tuber

Regions Of Roots





- **The main functions of roots are:**

i.) **Absorption** of water & minerals.
ii) **Anchorage** plants to soil.
iii) **Storage** of reserve food.
iv) Synthesis of **Plant hormones** or plant growth regulators (PGRs).

Modifications of Roots:

When roots deviates from these functions & do something else; they are said to be modified in their **Shape**, **structure** & **functions**.

Storage

Roots get swollen to store food

- **Tap roots** e.g., Carrot, Turnip
- **Adventitious roots** e.g., Asparagus, Sweet potato



Support

- Hanging root in Banyan tree. → **Prop roots**



- Stilt roots arising from lower nodes of stem → In Maize, sugarcane



Respiration

- In plants like **Rhizophora**, the roots grow upwards (negatively geotrophic) which are called **pneumatophores** & helps to get O_2 for respiration.



THE STEM

- It is ascending part of plant which arise from plumule of seed.
- It carries nodes and internodes (space between 2 nearest nodes)
  - ↳ Leaves arise from these places
- Stem also bear buds :

Axillary buds (many)





→ Terminal / apical bud (single)

↓

These buds modify themselves to make many tendrils (for support) or thorns (for protection)

Functions of Stem :

- Provide strength & support to whole plant.
- Helps in transport of H_2O, mineral, photosynthates.
- Bear branches where leaves, flowers, fruits are attached.
- Behave as an organ of reproduction : e.g., eyes of potato, stem of rose etc.

Modifications Of Stem:

1. For Perennation : Helping plant to tide over unfavourable conditions e.g., in Colocasia

2. For Storage : Underground stem of Zamikand (Ground + stem), Potato, Ginger, Turmeric and Colocasia

Turmeric



Colocasia



Potato



Ginger



3. For Support : Axillary bud modifies to become tendril ↓ thin, slender structures

e.g. Pumpkin, cucumber, watermelon, bitter gourd.



e.g. Pumpkin, cucumber, watermelon, bitter guard, grapewine

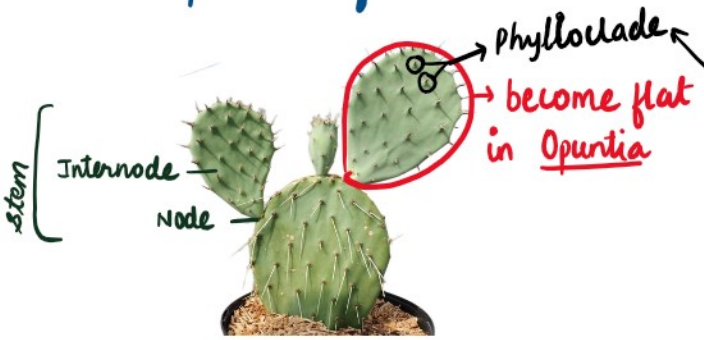


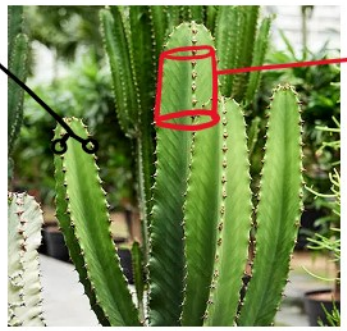
thin, slender structures

4. For **Protection**: These are pointed, woody structures formed by modification of axillary bud. (Thorns).
e.g. Bougainvillea & Citrus



5. For **Photosynthesis**: Stem modifies itself to do photosynthesis in plants of arid areas ($H_2O\downarrow$)





Fleshy, cylindrical in Euphorbia

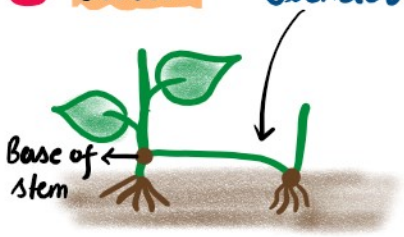
6. **Runner**: It helps grasses and strawberries to invade new areas (niches) and to regrow them (generation of new parts when old parts die) really fast.



7. **Stolon**: Slender, lateral branch grow aerially and touch

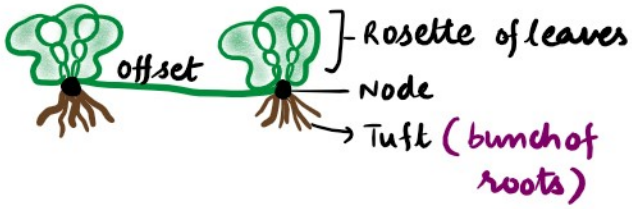
7. **Stolon** : Slender, lateral branch grow aerially and touch the ground.

e.g. Mint, Jasmine



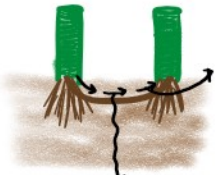
8. **Offset**: The internodes are small (∴ nodes are very close) in lateral branches of several aquatic plants like Pistia & Eichhornia (water hyacinth)

* In aquatic plants, offset helps in vegetative reproduction.



9. **Sucker** :

e.g. Pineapple, Chrysanthemum, Banana



Grow horizontally initially and come out obliquely (sloping)

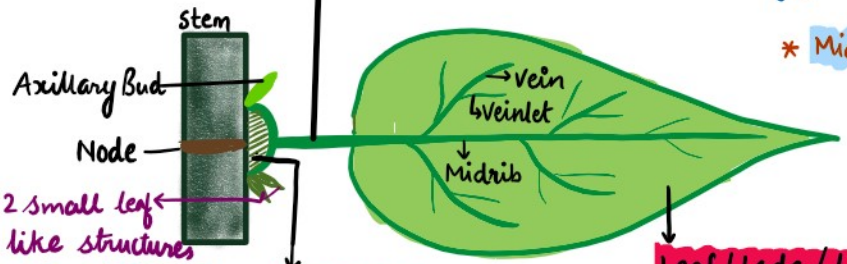
Lateral branch arising from underground part of stem

LEAF

- They are flat, green (Photosynthetic) structure arranged acropetally on stem branches.
- Leaves are part of shoot ∴ are formed from modification of cells of shoot apical meristem.

■ **Parts of leaf**: Leaf base, petiole, lamina

Petiole:
- Orient leaf towards light
- Help leaf to flutter (bring fresh air) ∴ Temp ↓



* Midrib, Veins, Veinlets: Provide rigidity and channel for transport

Leaf blade / Lamina:

Leaf base: Connect petiole to stem.
↳ It can show two types of modification:
(i) In monocots: It makes 'sheath' that covers the monocot stem partially or wholly.
(ii) In leguminous plants like pea or pulses, it swell & make 'Pulvinus'.

If present, plant is called **Stipulate** e.g., Pea

↓
It can be photosynthetic (as it is leaf like) and protective (can protect young developing leaf).

Venation:

- It is arrangement of vein & veinlets of leaf.
- It is of two types:

Reticulate

- Veins & Veinlets can intersect.

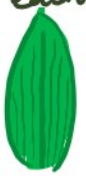


Intersection points
e.g., Dicots

- Dicots are usually tetra

Parallel

- Veins & Veinlets do not intersect with each other.



e.g., Monocots

- Monocots are usually trimerous

- Dicots are usually **tetra** or **pentamerous** means their floral parts are in multiple of 4 or 5.

- Monocots are usually **trimerous** i.e., their floral parts are in multiple of 3.

TYPES OF LEAVES

Simple leaf



No incision of leaf

Incisions are present but they do not touch the midrib

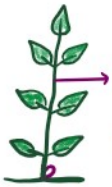
Compound leaf



- Axillary bud is not found on leaflets
- Incisions touch the midrib ∴ leaflets are formed

Pinnately Compound leaf

Have **pin**-like main midrib called rachis where leaflets are arranged.



Rachis

e.g. Neem

Palmately Compound leaf

All leaflets arise from a common point just like all fingers arise from **palm**.



e.g., Silk cotton

PHYLLOTAXY

It is the arrangement of leaves on branch of stem.

Alternate

- Leaves arise from alternate nodes ∴ **one leaf at a node**



Opposite

- **Two leaves** arise from **one node** in opposite direction.



Whorled

- **More than two** leaves arise from a node.





e.g., Mustard, Sunflower, Chinarose



e.g., Calotropis, Guava



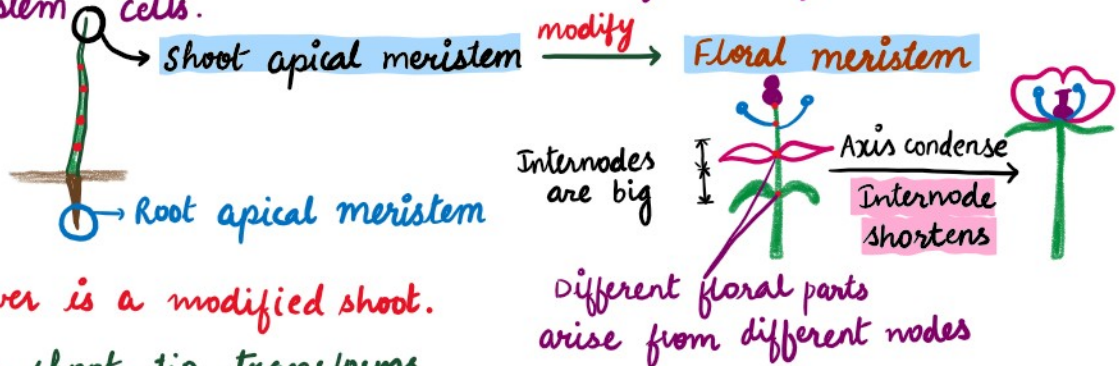
e.g., Alstonia

Modifications Of Leaves :

- Tendril : Green, climbing structure for support.
  e.g., Pea
- Spine : Pointed structure for defense or protection.
  e.g., Cactus (also reduce water loss).
- Storage : Fleshy leaves in onion and garlic.
- To catch insects : Pitcher of pitcher plant and trap of Venus fly trap are modified leaves.
- Petiole expands and make leaf like structure for photosynthesis : In Australian acasia, leaves are small in size and live for short time span only; thus petiole modify to take over their functions.

FLOWER:

- Flower is present on shoot ∴ is formed from shoot apical meristem cells.



∴ Flower is a modified shoot.

- When shoot tip transforms in a flower, it is always solitary (alone).

Structure Of flower

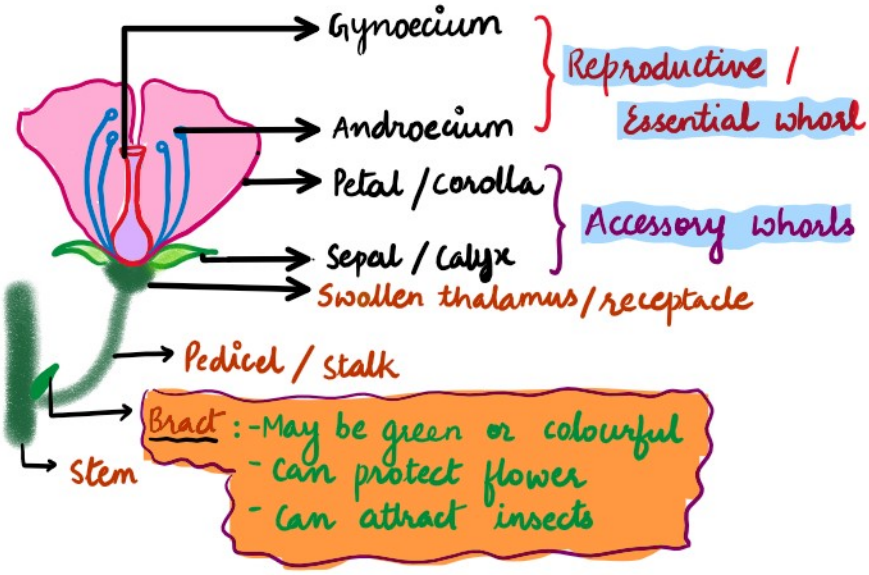
Structure of flower

- Outermost to innermost floral whorls

TRICK :

Kali Charan Aa Gaya

K C A G

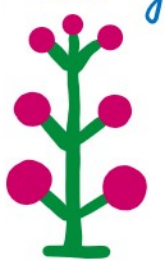


INFLORESCENCE:

- It is the arrangement of flowers on floral axis.
- It is of two types.

Racemose

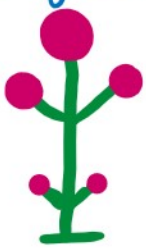
- Show acropetal arrangement i.e youngest flower is at tip.
- Unlimited growth as new flowers keep on growing as soon as axis grows.



e.g., Fabaceae

Cymose

- Show basipetal arrangement i.e. youngest flower at base.
- Tip contains biggest / oldest flower
  ∴ axis do not grow further.
  ∴ limited growth.



e.g., Solanaceae, Liliaceae

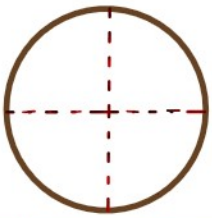
NOTE : · Inflorescence in Brassicaceae – Raceme 

- Asteraceae - Capitulum / head inflorescence
- Poaceae - Spikes of spikelet inflorescence

Symmetry of flowers:

Radial / Actinomorphic ⊕

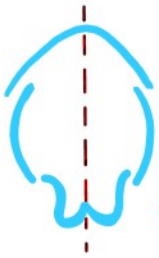
- Flowers can be cut into two equal halves at more than one axis.



e.g., Mustard, datura, chilli

Bilateral / Zygomorphic %

- Only one axis can divide a flower into two equal halves.



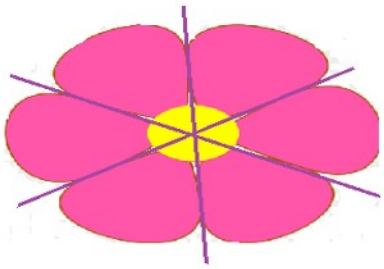
e.g., Pea, Gulmohar, Cassia, Bean

Asymmetrical

- No axis can divide flower into two equal halves



X e.g., Canna



Actinomorphic



Zygomorphic



Asymmetrical

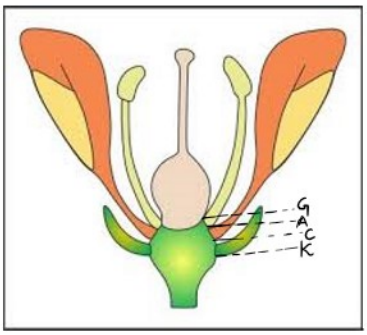
Symmetry of flower

Position of Calyx, Corolla and Androecium w.r.t. Gynoecium

Hypogynous Condition

below + gynoecium

∴ ovary is superior as all other parts are present below it.



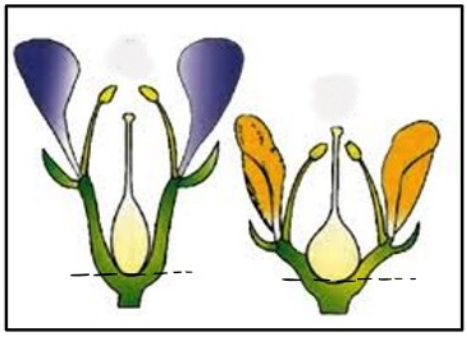
- Represented as : $\underline{G}$

e.g., Mustard, China rose, Brinjal

Perigynous Condition

Periphery

- All parts arise from the periphery of ovary from almost same level of ovary.
- Parts other than ovary arise from rim of thalamus.



- Represented as : $\overline{G}$

e.g., Plum, Rose, Peach

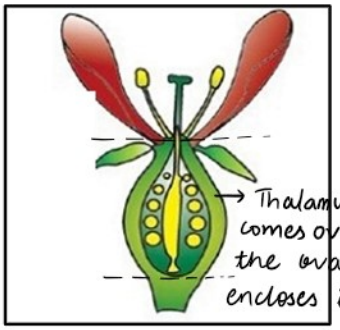
- Ovary looks inferior in diagram but its not ∴ called half inferior

Epigynous Condition

above

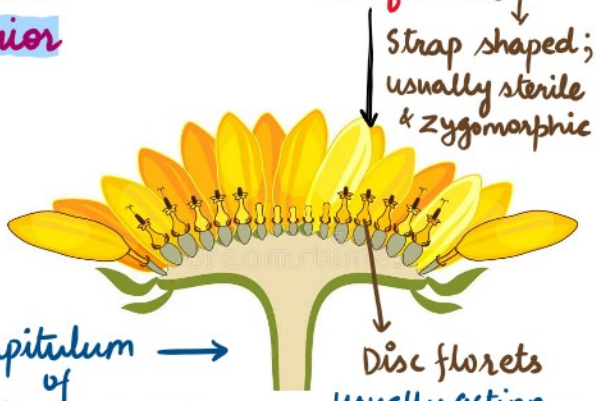
- All other parts are positioned above ovary.

∴ inferior ovary



- Represented as : $\overline{\underline{G}}$

e.g., Guava, cucumber, ray florets of sunflower



Calyx and Corolla



Corolla / Petals : have distinct shapes & colours to attract different insects (pollination)



shapes & colours to attract different insects (pollination)

- Petals can be :
  - free (polypetalous)
  - fused (gamopetalous)





Calyx / **Sepals** : Green in colour, protects the developing petals & inner floral parts

- Sepals can be :
  - free (Polysepalous)
  - fused (gamosepalous)





* Perianth (P) is a term used for family **Liliaceae** & **Poaceae** when calyx and corolla are indistinguishable.

Aestivation :

It is pattern of arrangement of one calyx w.r.t. all other calyx of a flower or arrangement of one corolla w.r.t. all other corolla in a flower.

1. Valvate :

e.g., Calotropis

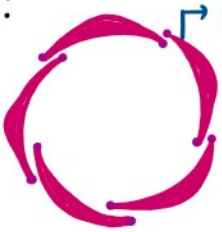


→ Sepals or petals just touch each other at margins **without overlapping**



2. Twisted :

e.g., Cotton, China rose, Lady's finger



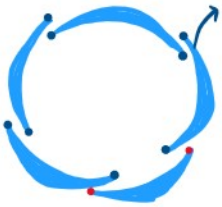
→ One margin of the appendage overlaps with that of the next one in **orderly way**.



overlapping edge of a petal

3. **Imbricate** :

e.g., Cassia, Gulmohar



Margins overlap each other but do not follow any order.



Both edges are overlapping

Both edges are underlapping

4. **Vexillary / Paplionaceous** : In family Fabaceae or Paplionaceae

e.g., Pea, bean



Largest petal called **standard / vexillum**

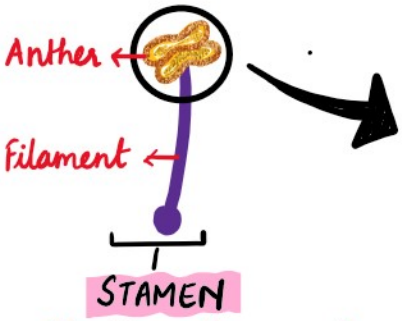
Lateral wing

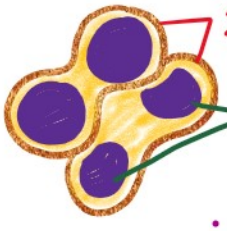
Anterior, fused **keels** (smallest)



Androecium :

- All stamens in a flower constitutes androecium.





2-lobes in anther ∴ it is **bilobed**

Two-theca (compartments) in every lobe

∴ **Dithecous** condition

∴ Total 4-pollen sacs (theca) are in anther

↓

Pollens are produced here

* Sterile stamen is called **'STAMINODE'** → can't reproduce

- **Stamen can be attached to** :

→ Petal

→ Tepal / Perianth

Petal → Epipetalous condition
e.g., Brinjal

Tepal / Perianth → Epitepalous or Epiphyllous
e.g., Lily

■ **Stamens can be :**

Free → Polyandrous condition

United — called Synandrous condition (i.e., Fused) to various degrees

- Monoadelphous (single Bundle)
- Diadelphous (Two Bundles)
- Polyadelphous (Many (>2) Bundles)



All stamens in one main bundle
e.g., China-rose



9 stamens together
one stamen seperated
e.g. Pea



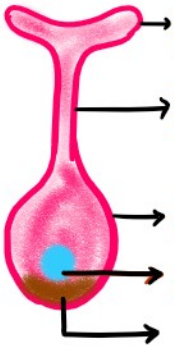
Many bundles
e.g., Citrus

NOTE: Sometimes, within the flower - the length of filaments is variable & thus, the length of stamen also varies i.e., in case of Salvia & Mustard.

Gynoecium :

- Represents the female reproductive part made of one or more carpels or pistils.

represents the female reproductive part made of one or more carpels or pistils.



Stigma : Receptive surface for pollens

Style : Long tube, connecting stigma to ovary

ovary : Basal, enlarged part of carpel

One or more ovules attached to placenta

Flat, cushion-like structure called **Placenta**

- If there is only one pistil in flower, it will obviously be free.
- If there are more than one pistils ; they may be :

Free → Apocarpous



e.g., Lotus, Rose

Fused → Syncarpous



e.g., Mustard, Tomato

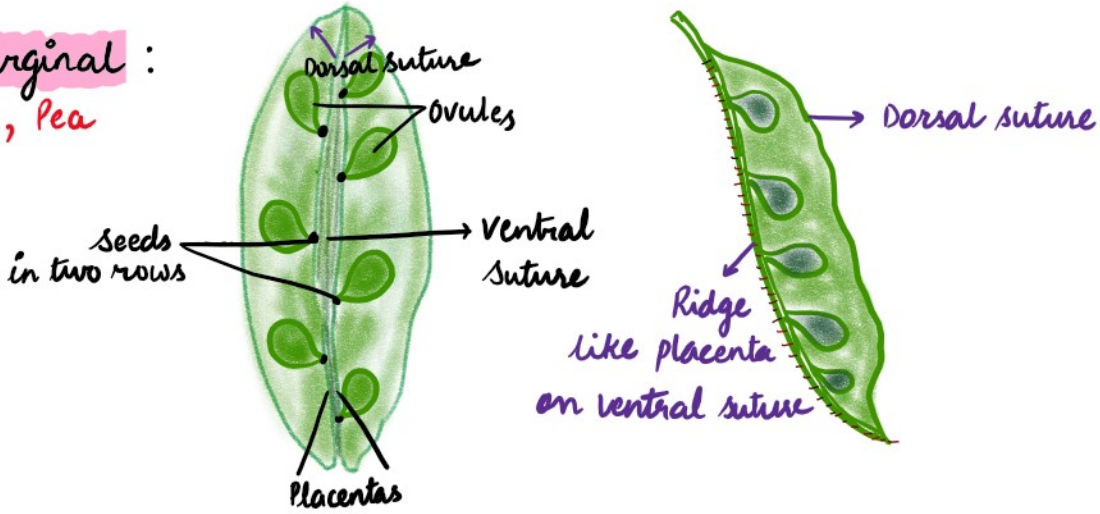
* After fertilisation : The **ovule** becomes **seed** and **ovary** becomes **fruit**.

PLACENTATION

- It is the arrangement of ovules on placenta of ovary.
- Arrangement of ovules in ovary is basically represented by arrangement of seeds in fruit.
- **Types of placentation** :

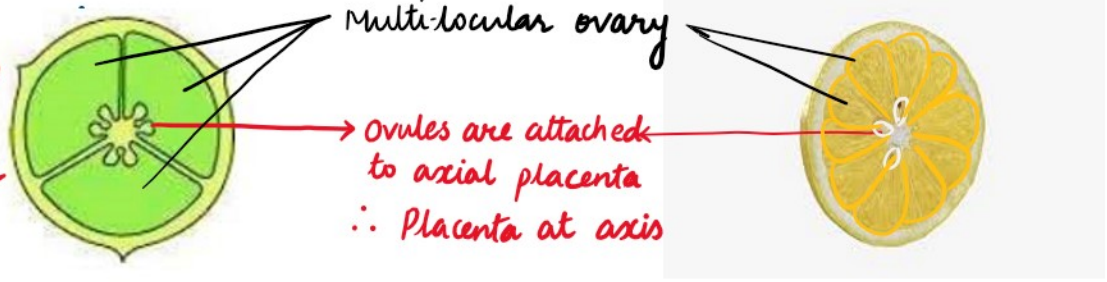
1. Marginal :

e.g., Pea



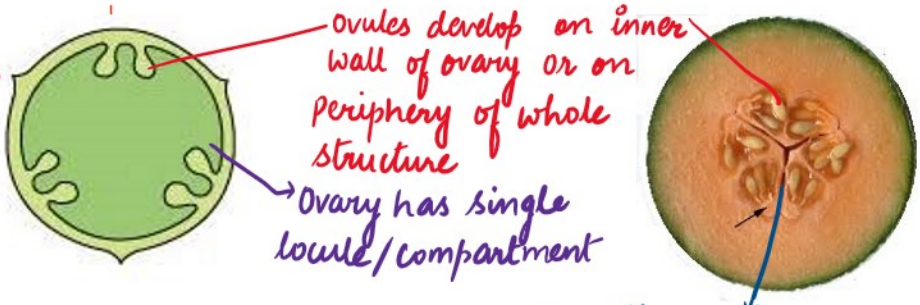
2. Axile :

e.g., Lemon, Tomato, china rose



3. Parietal :

e.g., Mustard, Argemone

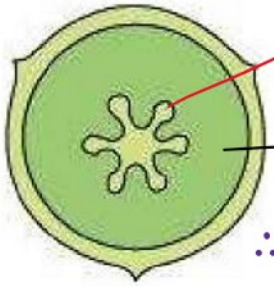


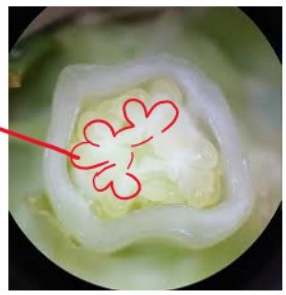
...locule / compartment

Sometimes ovary appear 2,3-chambered due to formation of false septa called 'Replum'

4. **Free-central** :

e.g., Dianthus
Primrose





5. **Basal** :

e.g. Sunflower, Marigold



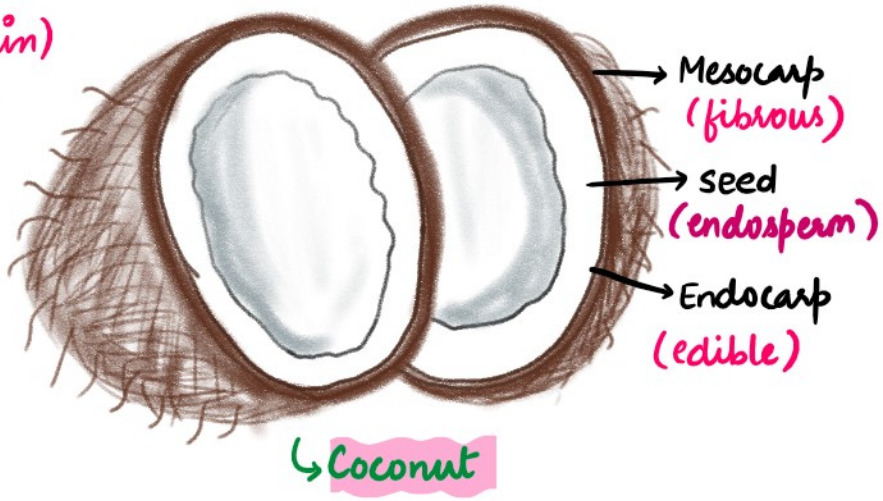


THE FRUIT:

- Angiosperms have some special features like flower, fruit and double fertilisation.
- Parthenocarpic fruits : Fruits formed without fertilisation.
  ∴ they are 'seedless'.
- Fruit consists of : i) Seeds
  ii) Wall / Pericarp : can be:
  - Fleshy → can be divided into outer epicarp, middle mesocarp & inner endocarp
  - Dry



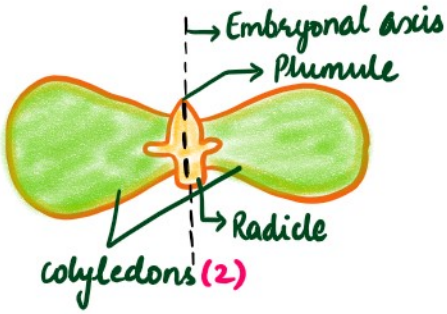




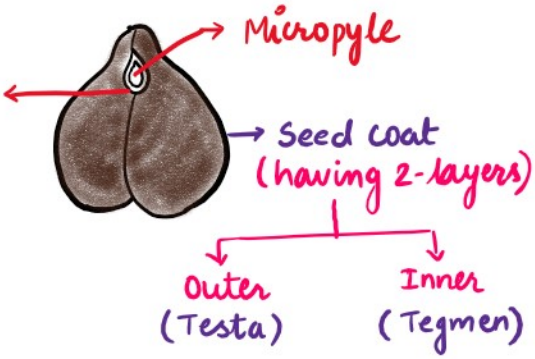
- In both mango & coconut, the fruit is drupe which is developed from monocarpellary superior ovary ($\underline{G_1}$) and are one seeded.

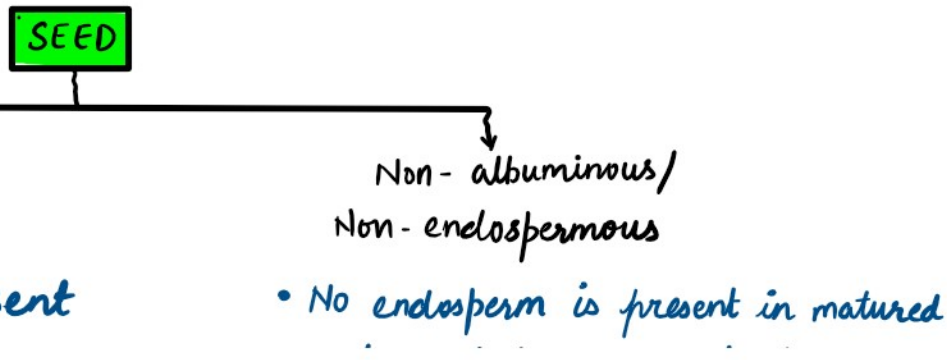
SEED :

- A typical seed is made of **embryo & seed coat**.
- Embryo is made of :- embryonal axis,
  - plumule,
  - radicle,
  - cotyledons.



A **scar on seed coat** through which the developing seed is connected to fruit





- Endosperm is **present** at maturity.
  i.e., Castor (dicot) & most monocots

- No endosperm is present in matured seed as it is consumed during embryonic development.
  e.g., most dicots (like pea, gram, bean) & orchid (monocot)

Monocot seed :

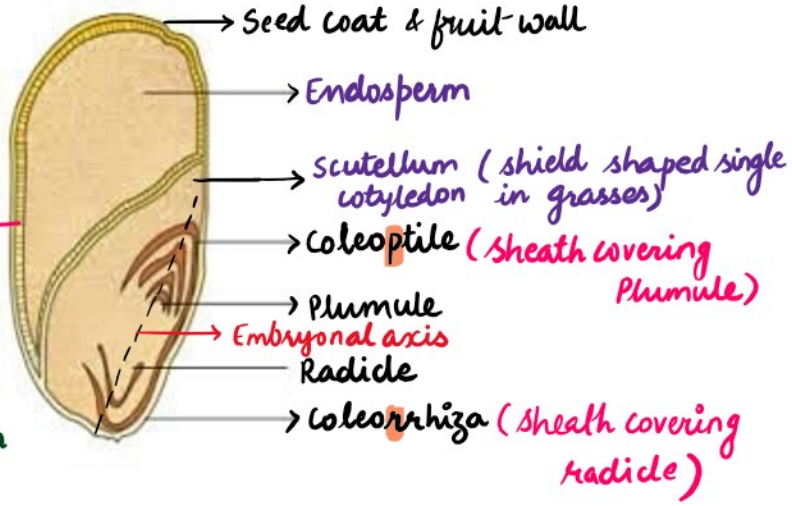
- In cereals like maize, the seed coat is membranous & it is fused with fruit wall.

Aleurone layer ←

↓

Aleuro means protein

∴ it is a layer of protein that separated embryo from endosperm



Semi-technical Description Of A Typical Angiosperm

- All crucial characters of plants needs to be described in a short & simple way using scientific symbols in a particular sequence.
- All characters are written in a particular sequence via describing plants like :
  1) Habit : Herb, Shrub or tree
  2) Vegetative characters : Root, stem, leaf
  3) Reproductive characters : Inflorescence, androecium, gynoecium

- Some symbols used to write floral formula:

K : Calyx
C : Corolla
P : Perianth / Tepals

A : Androecium
G : Gynoecium

$\underline{G}$: Superior ovary
$\overline{G}$: Inferior ovary
G- : Half inferior ovary

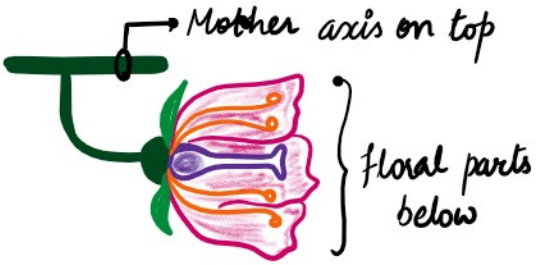
♂ : Male
♀ : Female
⚥ : Bisexual

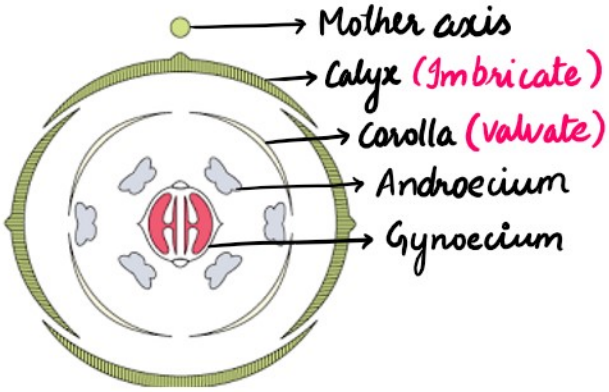
⊕ : Actinomorphic
% : zygomorphic

Br : Bracteate
EBr : Ebracteate

- Fusion is shown by bracket '( )', e.g., $A_{(9)+1}$ means '9' stamens are fused.
- Adhesion is shown by arc line drawn above floral parts e.g., C⌒A ; means petals & androecium are connected ∴ Epipetalous condition.

FAMILY : BRASSICACEAE / CRUCIFERAE





Floral formula : ⊕ ⚥ $K_{2+2}\ C_4\ A_{2+4}\ \underline{G}_{(2)}$

Floral formula : ⊕ ♂ $K_{2+2}\ C_4\ A_{2+4}\ \underline{G}_{(2)}$



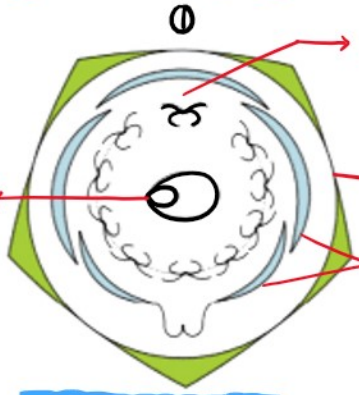
Floral diagram

⊕ : Radial symmetry
♂ : Bisexual flower
K_{2+2} : 2 - calyx having both edges overlapping & 2 - calyx having both edges underlapping
C_4 : All petals are similar ∴ 4 is written directly
A_{2+4} : 2 stamens are more distant (outer) & 4 are present inner to them (∴ 2+4).
$\underline{G}_{(2)}$: 2 - fused carpels (superior)

NOTE: Tetramerous family

- Inflorescence : Raceme
- Fruit : Siliqua, Silicula (generally non - endospermous seeds)
- e.g., Mustard, radish, turnip, candytuft etc.

FAMILY : FABACEAE / PAPLIONACEAE



1 stamen is separate and rest (9) are fused ∴ diadephous condition

Sepals are 5 & fused ∴ (5) : Valvate (Imbricate)

Vexillary aestivation ∴ 1 + 2 + (2)

Monocarpellary, Superior ovary ∴ $\underline{G}_1$

Floral diagram

Floral formula : % ♂ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_1$

- Members of family fabaceae are also called leguminous plants.

■ **Vegetative Characters** :

- Habit : Herbs, shrubs or trees
- Roots : have root nodules
- Stem : is erect or climber (tendrils)
- Leaves : Alternate; simple or pinnately compound; pulvinate; stipulate; retiulate venation (since it is dicot).

■ **Floral characters** :

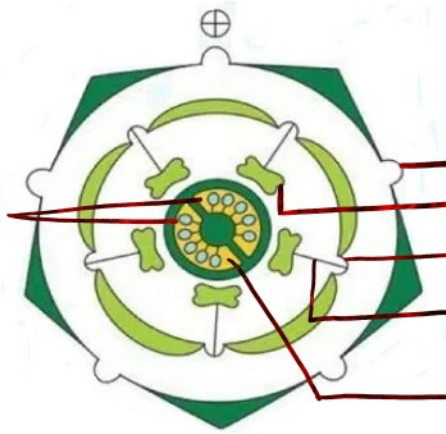
- Inflorescence : Racemose

Floral characters :

- Inflorescence : Racemose
- Calyx : Valvate or imbricate
- Androecium : Dithecous anther
- Gynoecium : Ovary with many ovules and single style.
- Fruit : Legume ; seeds are non-endospermic & one to many in numbers.

Examples : Sunhemp (Fibre), Sesbania (fodder), Lupin (ornamental), Trifolium (Fodder), Pea, Pulses, Soyabean, Groundnut (oil), Indigofera (dye), Mulaithi (Medicine)

FAMILY : SOLANACEAE (Potato family)



Obliquely placed septa ; 2-fused carpels ∴ $\underline{G}_{(2)}$

Sepals : 5-fused ; Valvate ∴ (5)
Stamens : 5-free
Petals : 5-fused ; Valvate ∴ (5)
Epipetalous : Stamens connected to petals
Placenta : Swollen with many ovules (axile placentation)

Floral diagram

Floral diagram : $\oplus\ \female\ K_{(5)}\ \overset{\frown}{C_{(5)}}\ A_5\ \underline{G}_{(2)}$

Vegetative characters :

- Habit : Herbs (mostly), shrubs or small trees (rare).
- Stem : Herbaceous, rarely woody ; aerial ; erect cylindrical ; branched ; solid or hollow ; hairy or glabrous (smooth) ; underground stem in potato.
- Leaves : Alternate ; simple or pinnately compound (rare) ; ex-stipulate ; reticulate venation (since it is dicot).

Floral characters :

- Inflorescence : Cymose, solitary, axillary
- Calyx : Sepals are persistant

- Inflorescence : Cymose, Solitary, axillary
- Calyx : Sepals are persistant
- Fruit : Berry or capsule
- Seed : Many ; endospermous

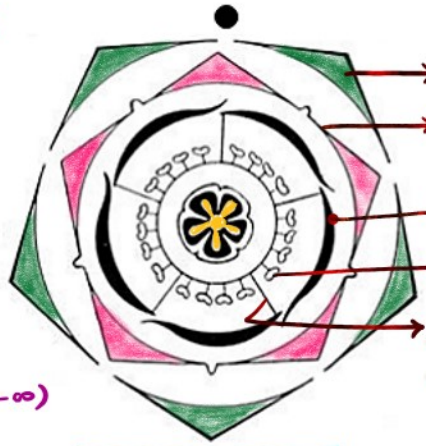
Example : Medicine (Belladona , Ashwagandha)
Ornamental (Petunia)
Fumigatory (Tobacco)
Vegetables (Tomato , Brinjal , Potato)
Spice (Chilli)

FAMILY : MALVACEAE

Examples : Hibiscus (china rose), cotton , Bhindi , Chorchorus (white jute), Balsa wood (lightest wood), Pavonia (Hina) ↓ scents

Floral formula :

$$Br \oplus \delta^{\circ} \; Epi_{3-7} \; K_{(5)} \; \overset{\frown}{C_5} \; A_{(\infty)} \; \underline{G}_{(5-\infty)}$$

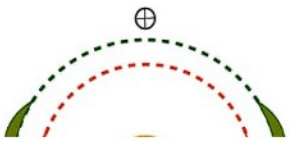


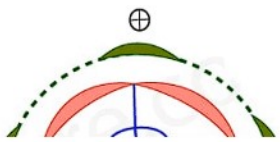
Floral diagram

- Inflorescence : generally Cymose
- Fruit : Loculicidal capsule
- Mucilage is present on flower or fruit.
- Thin and stellate (star) shaped hairs are present on stem & fruit.

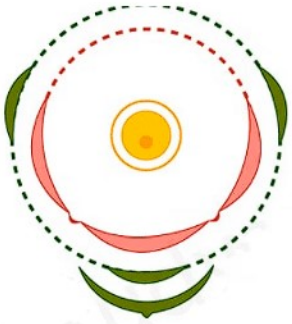
FAMILY : ASTERACEAE / COMPOSITAE

- Inflorescence is capitulum / head and fruit is cypsella.
- Examples: Garden lettuce, Sunflower, genda, bhringraj (medicinal), Artemisia (make santonin chemical which remove worms).

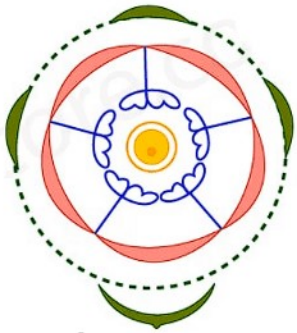




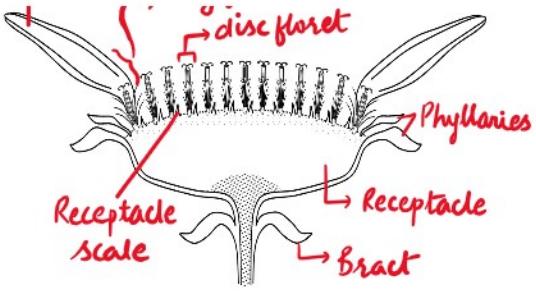




Ray - Floret



Disc - Floret



Capitulum / Head

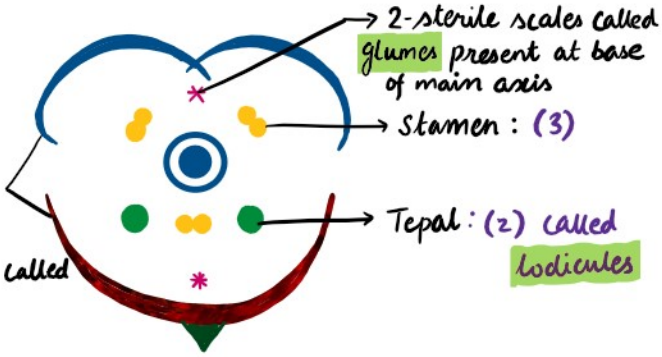
Br % ♀ $K_p C_{(5)} A_0 \underline{G}_{(2)}$

Br ⊕ ♂ $K_p C_{(5)} A_{(5)} \underline{G}_{(2)}$

FAMILY POACEAE / GRAMINAE

Examples: Cynodon (grass), Oat/Jai, Bajra, wheat, Jawar, maize, rice, barley, moonj (fiber), lemon grass, sugarcane

Superior & inferior bracts & bracteole called Palea



Floral diagram

Floral formula : ↓

Br % ♂ $P_2 A_3 \underline{G}_1$ or $\underline{G}_3$ → monocot family

- Inflorescence : Spike of spikelet
- Fruit : Caryopsis